

Morgan Patel

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EDUCATION

Ridgeway University | Ph.D. candidate in Statistics | Metro City, USA | Sep 2021 - Expected May 2026

Ridgeway University | B.S. in Mathematics | Metro City, USA | Sep 2016 - May 2020

EXPERIENCE

Northpeak Capital | Quantitative Research Intern | Metro City, USA | Jun 2025 - Aug 2025

- Built a short-horizon order-book imbalance signal for liquid index futures that added 0.4 Sharpe to the desk's book over 18 months of out-of-sample backtest after costs.
- Cut the signal's daily turnover from 34% to 21% with a cost-aware position smoother, keeping 90% of gross returns and lowering estimated slippage by a third.
- Validated the signal on 6 years of tick data with purged walk-forward splits and an embargo period, so no look-ahead information leaked into training.
- Ran the signal through purged walk-forward validation on six years of tick data, with an embargo so nothing from the future leaked in.
- Presented the signal, its capacity estimate and failure cases to the portfolio managers, who approved a small live allocation for the next quarter.
- Documented the backtest assumptions, transaction-cost model and known failure regimes in the desk's research wiki for future interns.

Ridgeway University | Graduate Research Assistant, Statistical Learning Lab | Metro City, USA | Jun 2020 - Aug 2021

- Owned the lab's simulation pipeline on the shared cluster and the reproducibility checks other students relied on.
- Derived a variance bound for a sparse regression estimator; my proof tightens the previous bound by a log factor and is now Section 3 of a paper under review at JASA.
- Taught weekly recitations for 60 students in graduate probability, writing 12 problem sets and earning a 4.8/5 teaching rating.
- Released an open-source R package for high-dimensional covariance estimation while maintaining the lab's shared cluster, organizing the weekly reading group and grading for two courses, which was downloaded 3,000 times in its first year.

PROJECTS

Volatility Forecasting Study | Independent Research | Python, PyTorch | Jan 2024 - Present

- Pioneered a data-driven, research-first approach to volatility modeling that delivered robust, actionable insights.
- Cut forecast error from 0.20 to 0.15, a 33% improvement, by adding realized-volatility features and an asymmetric loss.
- Ran the robustness study on the VX9 backtest harness and wrote up the results as a 12-page working paper.

Kaggle Market Prediction Competition | Team of 3 | Python | Mar 2023 - Jun 2023

- Collaborated with teammates across multiple stages of the competition, contributing to the team's overall results.
- Validation leakage was identified and reduced through the use of time-grouped folds, and the score gap was closed.
- Wrote the team's feature-selection script using permutation importance, trimming 900 candidate features to 300 without losing validation score.

SKILLS

Programming: Python, R, SQL, C++, Kafka

Methods: time-series econometrics, Bayesian inference, gradient boosting, PyTorch